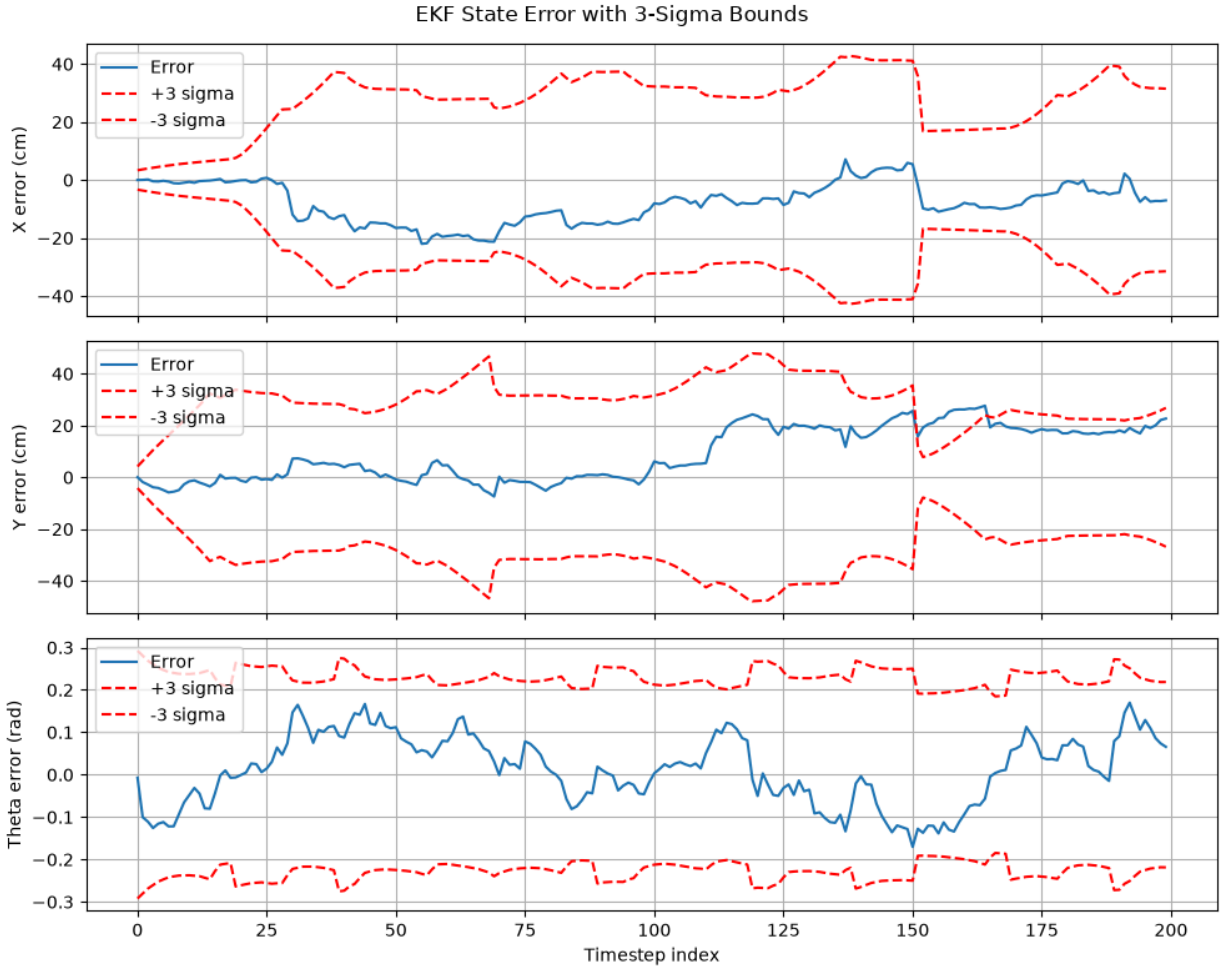
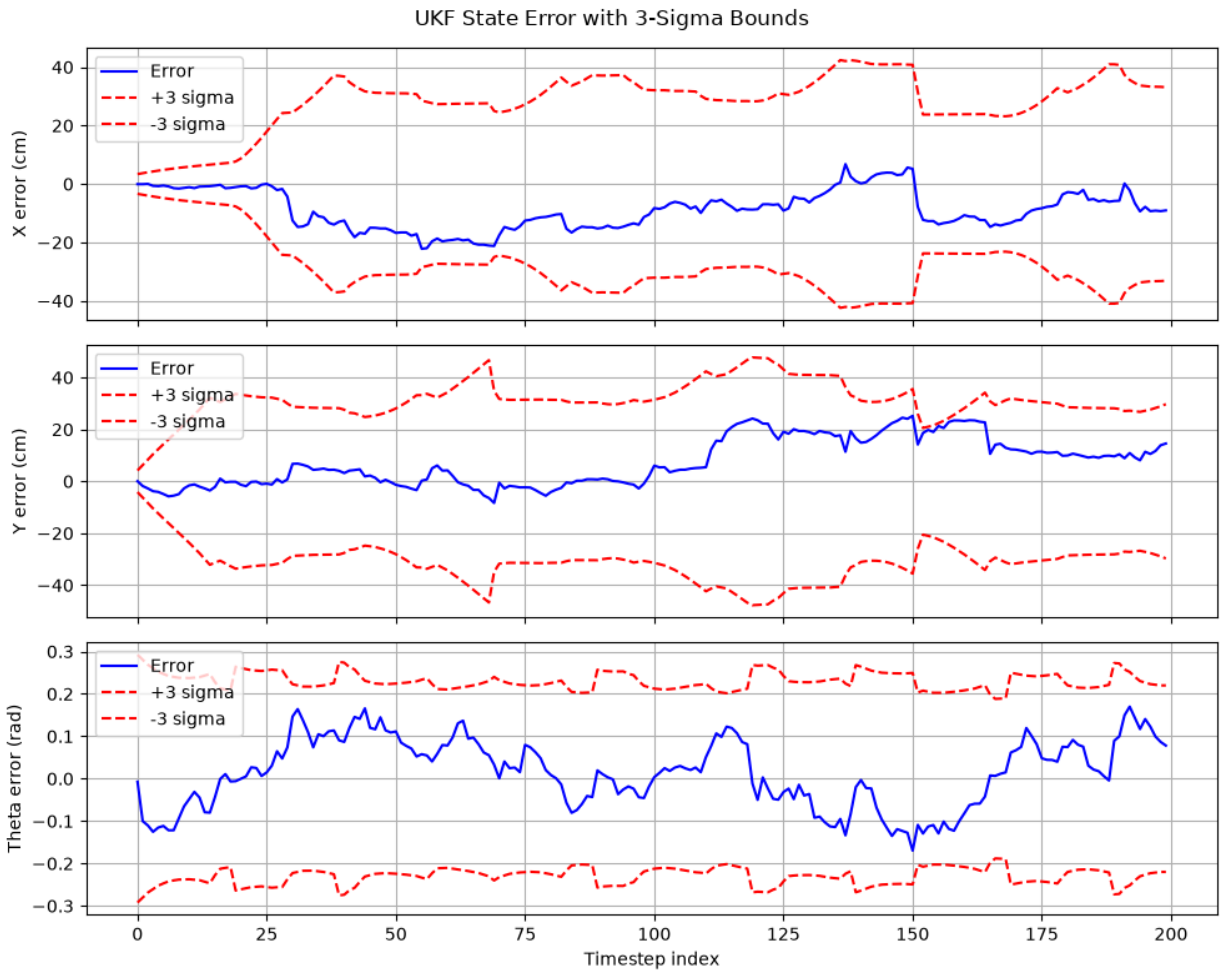
Lab 4: EKF and UKF Mapping
Austin J. Erickson

1. EKF and UKF localization was completed
    a. Video submissions included in .zip folder
2. Plots of error poses for UKF and EKF, along with terminal output of how often the variables stay within 3-sigma bounds (all done using -d data/data.npz given dataset)

NOTE: all calculations of algorithms was done using alpha=1 and beta=2 values



Fraction of steps with |X error| <= 3 sigma: 100.00%
Fraction of steps with |Y error| <= 3 sigma: 93.00%
Fraction of steps with |Theta error| <= 3 sigma: 100.00%



Fraction of steps with |X error| <= 3 sigma (UKF): 100.00%
Fraction of steps with |Y error| <= 3 sigma (UKF): 100.00%
Fraction of steps with |Theta error| <= 3 sigma (UKF): 100.00%

3. Investigate the properties of the filters:
    a. As the sensor or motion noise approaches zero, the error on both the UKF and EKF also shrinks proportionally. This also makes the two algorithms more closely resemble each other, most likely due to the linearization in the EKF becoming more accurate. Notably, setting the noise to zero causes the UKF to crash, due to the cholesky decomposition requiring a positive definite matrix, whereas the EKF runs, but returns NaN
    b. When the noise is underestimated, the filters become “overconfident,” meaning the actual error does not change much, but the ellipse is smaller, making the truth fall outside of the estimated ellipse far more. When the noise is overestimated, the filter becomes too conservative, making the ellipse is much larger, and the measurements given are underused, resulting in poor estimations.

Apologies, no extra credit was done for this assignment, that UKF filter really took it out of me for a bit there